

# Analysis First-Year Exam, August 2020, Part 1

---

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let  $(a_n)$  and  $(b_n)$  be bounded real sequences. Define  $\liminf a_n$  and  $\limsup b_n$ . Prove that

$$\liminf(a_n + b_n) \leq \liminf a_n + \limsup b_n \leq \limsup(a_n + b_n).$$

2. Let  $A$  be a subset of the metric space  $M$ . For each  $t > 0$  let

$$(A)_t = \{x \in M : (\exists a \in A) \, d(x, a) < t\}$$

$$[A]_t = \{x \in M : (\exists a \in A) \, d(x, a) \leq t\}.$$

Is  $\bigcap_{t>0} (A)_t$  equal to the closure of  $A$ ? Is  $\bigcap_{t>0} [A]_t$  equal to the closure of  $A$ ? Proof or counterexample, as appropriate.

3. Let  $f : [0, 1] \rightarrow [0, 1]$  be continuous and assume that for each positive integer  $n$  there exists  $t \in [0, 1]$  such that  $|f(t) - t| < 1/n$ . Prove that there exists  $a \in [0, 1]$  such that  $f(a) = a$ .

4. Let  $f : X \rightarrow Y$  be a map between metric spaces, with graph

$$G = \{(x, y) : y = f(x)\} \subseteq X \times Y.$$

Prove that  $f$  is continuous if  $G$  is compact.

5. The function  $f : (-1, 1) \rightarrow \mathbb{R}$  is differentiable and its derivative is bounded. Prove that there is a continuous function  $F : [-1, 1] \rightarrow \mathbb{R}$  with the property that  $f = F|_{(-1, 1)}$ .